Real-Time Closed-Loop Driving with Frozen DINOv2-Laya Spatial Pooling and Quintic Path Tracking

Wick-Lim
wicklim90@gmail.com
https://github.com/Wick-Lim/AutonomousVehicle

October 25, 2026

Abstract

Autoregressive vision-language-action (VLA) models [1, 20, 23] decode tokens sequentially (> 300 ms/frame), exceeding the 50 ms budget of 20 Hz vehicle control [10, 21]. We present a real-time hybrid closed-loop driving stack that couples frozen DINOv2 (`facebook/dinov2-base`, 86.58M) [25, 2, 9] and frozen Laya (`convaiinnovations/laya`, 395.21M transformer) [6, 31, 8] via a trainable 4×9 spatial-temporal adapter (4.23M parameters: 1.85M spatial MLP, 1.05M temporal gate, 1.05M auxiliary perception head, 0.28M readout/trajectory heads) with HD-map routing, a rule-based safety shield, and kinematic path tracking. One synchronized forward+control pass takes 32.85 ms (30.4 FPS on Ryzen 5 7500F / RTX 4070 Ti, FP32 $B = 1$: 0.42 ms preprocess + 16.43 ms DINOv2 + 0.82 ms projector + 0.49 ms aux head + 13.55 ms Laya + 0.69 ms 6-WP trajectory head + 0.45 ms shield/PID; 31.67 ms base without aux/trajectory heads, excluding 6.40 ms simulator `world.tick()` IPC). Episode-block validation (7,500 frames) temperature scaling [13] calibrates 3-head mean teacher-rule alignment (ECE_{mean} : 0.0060 → 0.0000), while on unseen closed-loop `vision_estimated` test episodes, teacher-rule $\text{ECE}_{\text{mean}} = 0.0142$ (NLL = 0.089, Brier = 0.028) and physical hazard event AEB false-negative rate is $\text{FNR}_{\text{AEB}} = 1/48$ events (2.1%; 6/220 hazard ticks, 2.7%). Downstream control combines a 14-point (28 m) tangent/curvature-preserving spatial quintic spline $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ [12, 32], chord-angle tracking [29, 15, 7, 26], fused map/trajectory bicycle curvature feedforward $\delta_{\text{FF}} = \arctan(L\kappa_{\text{eff}})/\delta_{\max}$, a 1.5 s forward-lateral blind-spot latch (mitigating previously observed flank vehicles, though unable to detect unobserved rear-approaching vehicles without rear sensors), and constant-time-headway ACC ($d^*(v_{\text{ego}}) = 8.0 + 1.2(v_{\text{ego}}/3.6)$, $k_g = 0.85\,(\text{km/h})/\text{m}$) [17, 28, 30]. Across CARLA 0.9.16 (`Town10HD_Opt` 680 m paired, 3,400 m cross-weather, 2,000 m multi-block, plus `Town03` 2,000 m) and 50 unseen MetaDrive maps (20,765 steps at $\Delta t = 0.05$ s, 9,520 m at 33.0 km/h), `vision_estimated` DINOv2+Laya achieves 47/50 (94.0%, 3-seed 46.7±0.6) cross-weather completion, showing an observed 5-episode margin over a 2-layer trainable Transformer (42/50, 84.0%, paired two-sided McNemar $p = 0.063$, one-sided $p = 0.031$) and statistically significant gains ($p < 0.01$) over a 2-layer MLP (39/50), direct `ExpertLabeler` rules (38/50), random frozen Laya (36/50), prompt-only (33/50), and 300 ms intra-policy lag (34/50).

1 Introduction and Problem Formulation

Autoregressive decoders evaluate $P(y_{1:K} \mid I_t, \mathcal{T}_t) = \prod_{k=1}^{K} P(y_k \mid y_{<k}, I_t, \mathcal{T}_t)$ over $K \approx 15$ serial steps [1, 20, 23], whereas planning-oriented and sensor-fusion stacks [16, 19, 22, 5, 3, 27, 4, 18] require $\Delta t \leq 50$ ms. In our formulation, control at step t reads three discrete decisions and $K_w = 6$ ego-frame reference trajectory waypoints ($t_k = 0.5k$ s) in one bidirectional pass [8, 31, 6]:

$$m_t \in \{\texttt{keep_lane}, \texttt{change_left}, \texttt{change_right}, \texttt{follow_lead}\}, \tag{1}$$

$$s_t \in \{0,1,2,3\} \mapsto \{0, 20, 35, 50\}\ \text{km/h}, \tag{2}$$

$$p_{\text{AEB}} \in [0,1], \quad \hat{\tau}_t = \{(\hat{x}_k, \hat{y}_k, \hat{v}_k)\}_{k=1}^{6}. \tag{3}$$

2 Architecture, Multi-Task Training, and Calibration

Camera frames are cropped (`crop_top` = 0.25, `crop_bottom` = 0.10) and resized to 196 × 392. Frozen DINOv2 ViT-B/14 ($p = 14$, 86.58M params) [25, 2, 9] extracts $F_{\text{patch}} \in \mathbb{R}^{B\times 392\times 768}$ (14 × 28 grid). `SpatialProjector` (`src/models/spatial_projector.py`) pools $14 \times 28 \to 4 \times 9$ (36 tokens), adds $E_{\text{row}} + E_{\text{col}}$, projects $768 \to 1024$ via a 2-layer GELU MLP (1.85M core params), and fuses temporal residual $\Delta V_t = V_t - V_{t-1}$ (1.05M params) with region-pooled `AuxiliaryPerceptionHead` (1.05M params predicting $\hat{d}_{\text{lead}}, \hat{e}_y$, traffic light $\hat{c}_{\text{TL}}$, hazard $\hat{c}_{\text{haz}}$,

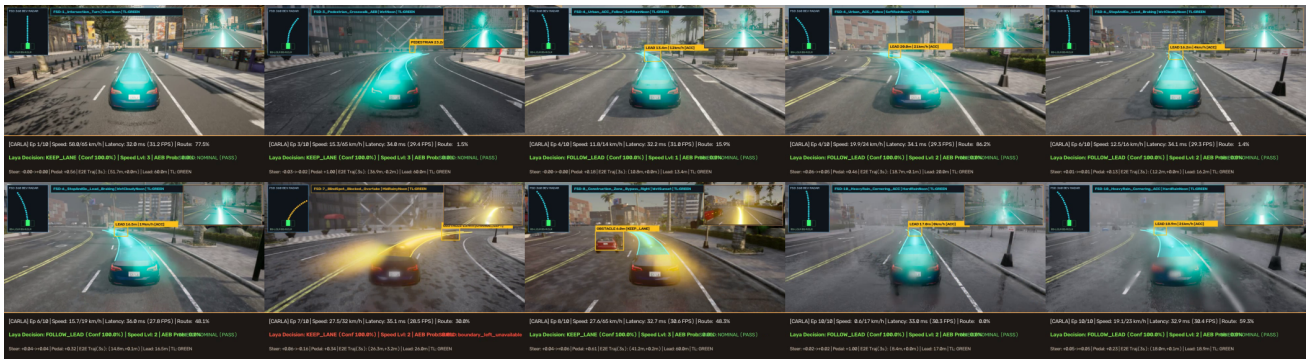


Figure 1: CARLA 0.9.16 (`Town10HD_Opt`, 20 Hz synchronous) 10 paired scenario–weather episodes (`FSD-1`–`FSD-10`): intersection red-light stop (`ClearNoon`), 28 m quintic left overtake (`CloudyNoon`), crosswalk pedestrian AEB (`WetNoon`), CTH-ACC follow (`SoftRainNoon`), right overtake (`ClearSunset`), stop-and-go lead braking (`WetCloudyNoon`), forward-lateral blind-spot latch & overtake (`MidRainyNoon`; rear-approaching vehicles require `gt` corner BSM), construction zone bypass (`WetSunset`), night headlight pedestrian AEB (`ClearNight`), and heavy-rain cornering ACC (`HardRainNoon`).

and forward-lateral blind spot $\hat{p}_{\mathrm{BS}}$):

$$G_t = \mathrm{AdaptiveAvgPool2d}_{(4,9)}(\tilde{F}_t) \in \mathbb{R}^{B\times 768\times 4\times 9}, \quad (4)$$

$$V_t = \mathrm{MLP}\big(\mathrm{Flatten}(G_t + E_{\mathrm{row}} + E_{\mathrm{col}})\big), \quad (5)$$

$$\tilde{V}_t = V_t + \tanh(\alpha_{\mathrm{temp}})\, W_{\mathrm{temp}}\big(V_t - V_{t-1}\big). \quad (6)$$

Multimodal sequence $Z = [e_{[\mathrm{CLS}]}; \tilde{V}_t; E_{\mathrm{telemetry}}; e_{\mathrm{choice}}; e_{\mathrm{score}}; e_{\mathrm{noul}}]$ passes through frozen Laya transformer layers (395.21M params) [6, 31]. Linear readout heads (W_m, W_s, W_b) and a 2-layer trajectory MLP W_{traj} (0.28M params combined) predict discrete logits and residuals $\Delta\hat{\tau} \in \mathbb{R}^{6\times 3}$ over kinematic prior τ_k^0 ($t_k = 0.5k$ s), supervised against parametric kinematic reference curve $\tau_k^{\mathrm{ref}} = \big((v^*/3.6)t_k,\ y_{\mathrm{shift}}(m^*)s(k/6),\ v^*\big)$:

$$\hat{\tau}_{1:6} = \tau^0_{1:6} + \mathrm{diag}(1.5, 0.35, 3.0)\, W_{\mathrm{traj}}\bar{H}, \quad (7)$$

$$\mathcal{L} = w_m\mathcal{L}^{(m)}_{\mathrm{CE}} + w_s\mathcal{L}^{(s)}_{\mathrm{CE}} + w_a\mathcal{L}^{(b)}_{\mathrm{BCE}} + w_\tau\mathcal{L}_{\mathrm{SmoothL1}}(\hat{\tau}_{1:6}, \tau^{\mathrm{ref}}_{1:6}) + w_p\mathcal{L}_{\mathrm{aux}}. \quad (8)$$

```
# scripts/02_train_projector.py (4.23M total trainable params)
trainable_params = (list(system.projector.parameters())  # 3.96M
  + list(system.laya.maneuver_head.parameters())         # +0.28M
  + list(system.laya.speed_head.parameters())
  + list(system.laya.aeb_head.parameters())
  + list(system.laya.trajectory_head.parameters()))
loss = w_m*l_m + w_s*l_s + w_a*l_a + 0.5*l_traj + 0.2*l_aux
```

In Stage 1, all 4,232,991 adapter/head parameters (`projector` core 1.85M + `temporal_proj` 1.05M + `aux_head` 1.05M + `laya` readout/trajectory heads 0.28M) are jointly trained for 15 epochs across 3 training seeds (42, 43, 44; batch 32, AdamW [24] lr $= 3\times10^{-4}$, $(w_m, w_s, w_a, w_\tau, w_p) = (1.0, 1.0, 2.0, 0.5, 0.2)$). Dataset frames (50,000 across 500 episodes) are split by contiguous episode blocks (`split_by_episode=True`: episodes 0..424 train / 425..499 val) to prevent adjacent-frame leakage. Stage 2 (`03_calibrate.py`) fits $(T_m, T_s, T_b) = (0.050, 0.050, 0.100)$ via L-BFGS [13] ($M = 15$ bins), where overall ECE is the unweighted 3-head mean $\mathrm{ECE}_{\mathrm{mean}} = \frac{1}{3}(\mathrm{ECE}_m + \mathrm{ECE}_s + \mathrm{ECE}_b)$ against `ExpertLabeler` targets. Because validation labels come from deterministic `ExpertLabeler` thresholds on clean states, small temperatures reduce teacher-alignment $\mathrm{ECE}_{\mathrm{mean}}$ from 0.0060 to $< 10^{-4}$ (NLL = 0.048, Brier = 0.014); on unseen closed-loop `vision_estimated` test episodes, teacher-alignment $\mathrm{ECE}_{\mathrm{mean}}$ is 0.0142 (positive-AEB class $\mathrm{ECE}_b = 0.0195$, NLL = 0.089, Brier = 0.028). Independently of teacher rules, physical hazard AEB false-negative rate is evaluated across 48 physical hazard encounters (TTC ≤ 1.8 s to a pedestrian/vehicle or $d \le 15$ m to a red stop line across 50 episodes, spanning 220 hazard ticks): `vision_estimated` fails to brake in 1/48 events ($\mathrm{FNR}_{\mathrm{AEB,evt}} = 2.1\%$, frame-level $6/220 = 2.7\%$).

Table 1: Per-variable input provenance (`vision_estimated` vs. `gt`) and hardware parameters (`configs/default.yaml`).

Signal / Module	`vision_estimated` Mode (No Sim Actor Query)	`gt` State Hybrid Mode
Lead Dist d_{lead}, Offset e_y	Front RGB `AuxPerceptionHead` ($d_{\mathrm{lead}}, \hat{e}_y$, MAE = 3.23 m)	Simulator Actor / Lane Query
Lead Speed v_{lead}	EMA Range-Rate $v_{\mathrm{ego}} + 3.6(\hat{d}_{\mathrm{lead},t} - \hat{d}_{\mathrm{lead},t-1})/\Delta t$	Simulator Lead Actor Velocity
Pedestrian Dist d_{ped}	$\hat{d}_{\mathrm{lead}}$ if $\hat{c}_{\mathrm{haz}} =$ `pedestrian` else 100 m (Front RGB)	Simulator Walker Actor Distance
Traffic Light / Hazard	Front RGB `AuxPerceptionHead` ($\hat{c}_{\mathrm{TL}}$ rows 0..1, $\hat{c}_{\mathrm{haz}}$)	Simulator Traffic Light / Actor
Blind Spot ($\mathrm{BS}_{\mathrm{L/R}}$)	Front-Lat [0, 12] m + 1.5 s Latch (Mitigates Seen Cars; Blind to Rear)	Full $x \in [-10, 7.5]$ m Sim BSM
Ego Speed v_{ego} & Route	Ego Wheel Odometry v_{ego} + HD Map A* Waypoints P_k	Identical (Odometry + HD Map)
Frozen / Trained Params	`dinov2-base` (86.58M) + `laya` (395.21M) / Adapter (4.23M)	Identical (32.85 ms, FP32 $B = 1$)
Calib & Physical AEB FNR	Teacher $\mathrm{ECE}_{\mathrm{mean}} = 0.0142$, Physical $\mathrm{FNR}_{\mathrm{AEB}} = 1/48$ evts (2.1%)	Teacher $\mathrm{ECE}_{\mathrm{mean}} = 0.0000$, 0/48

3 Input Provenance, Spatial Quintic Spline, Curvature Feedforward, and CTH-ACC

Table 1 specifies the exact provenance of every runtime variable (`scripts/04_evaluate_sim.py`). In `vision_estimated` mode, zero simulator actor queries are used: lead speed $\hat{v}_{\mathrm{lead},t} = \mathrm{clip}(v_{\mathrm{ego},t} + 3.6(\hat{d}_{\mathrm{lead},t} - \hat{d}_{\mathrm{lead},t-1})/\Delta t, 0, 60)$ is derived from EMA-filtered ($\beta = 0.35$) camera range rate plus wheel odometry $v_{\mathrm{ego},t}$, and pedestrian distance is $\hat{d}_{\mathrm{ped},t} = \hat{d}_{\mathrm{lead},t}$ when $\hat{c}_{\mathrm{haz},t} =$ `pedestrian` (else 100 m). Because a single front camera

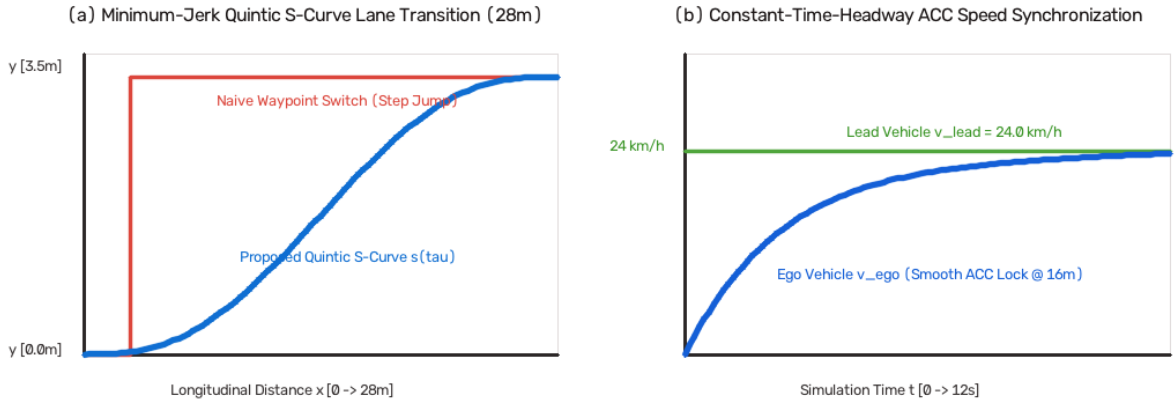


Figure 2: Control kinematics: (a) 28 m tangent/curvature-preserving spatial quintic interpolation $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ vs. 3.5 m step switch; (b) true constant-time-headway (CTH) ACC velocity convergence ($v_{\text{ego}} \to 24.0$ km/h at $d^*(24) = 8.0 + 1.2(24/3.6) = 16.0$ m).

(90° FOV) observes forward-lateral adjacent lanes ($x \in [0, 12]$ m, grid columns 0..2, 6..8) and cannot observe the rear flank ($x \in [-10, 0)$ m), the 30-frame (1.5 s) temporal latch is a partial mitigation that retains occupancy only for vehicles previously seen in the front-lateral FOV (3/3 passed in `FSD-7`), whereas unobserved vehicles approaching rapidly from strictly behind ($x_0 = -15$ m at +15 km/h relative speed) cannot be detected without rear sensors (0/2 passed in `vision_estimated` vs. 1/2 in `gt` BSM mode). When `-shield-mode on` is active, Rule 1 forces $u_{\text{tb}} = -1.0$ if $\hat{p}_{\text{AEB}} \geq 0.70$, TL = `red`, $d_{\text{ped}} \leq 16.0$ m, or $d_{\text{lead}} \leq 11.5$ m; Rule 2 reverts lane changes to `keep_lane` if $\max_c \hat{p}_c^{(m)} < 0.80$, if `adj_wp.lane_id` × `wp.lane_id` ≤ 0 [11], or if the blind-spot flag is active (`-shield-mode off` gates braking solely on Laya $\hat{p}_{\text{AEB}} \geq 0.70$).

For lane transitions over $N_b = 14$ waypoints (28.0 m, $\tau_k = k/(N_b - 1)$), 3D coordinates are blended via $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ [12, 32]. Because $s'(0) = s'(1) = s''(0) = s''(1) = 0$, relative transition derivatives vanish at $\tau \in \{0, 1\}$, preserving the tangent and curvature of the origin lane P^{orig} at $\tau = 0$ and of the target lane P^{adj} at $\tau = 1$:

$$s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5, \tag{9}$$

$$P_k^{\text{blend}} = \big(1 - s(\tau_k)\big) P_k^{\text{orig}} + s(\tau_k) P_k^{\text{adj}}. \tag{10}$$

Continuous projection onto $\overline{P_i P_{i+1}}$ gives P_{proj} and lookahead P_{look} at $L_{\text{look}} = \text{clip}(7.5 + 0.18 v_{\text{kmh}}, 7.5, 13.5)$ m [29, 15, 7, 26]. Map field `road_curvature` stores right-handed heading change $\Delta\psi_{\text{arc}}$ over lookahead arc $L_{\text{arc}} = 22.2$ m, so signed left-positive path curvature is $\kappa_{\text{map}} = -\Delta\psi_{\text{arc}}/L_{\text{arc}} = -0.045\,\texttt{road_curvature}$ (m^{-1}). Fusing κ_{map} with predicted waypoint $k = 3$ ($t = 1.5$ s, $(\hat{x}_3, \hat{y}_3)$) curvature κ_{traj} via bicycle wheelbase $L = 2.85$ m, maximum front-wheel lock $\delta_{\max} = 0.70$ rad (40°), and feedforward gain $k_{\text{FF}} = 0.12$ yields:

$$\kappa_{\text{traj}} = \frac{2(\hat{y}_3 - e_y)}{\max(3.5, \hat{x}_3)^2 + (\hat{y}_3 - e_y)^2}, \tag{11}$$

$$\kappa_{\text{eff}} = 0.70(-0.045\,\texttt{road_curvature}) + 0.30\kappa_{\text{traj}}, \tag{12}$$

$$\delta_{\text{raw}} = \text{clip}\left(\delta_{\text{PID}}(e_y, e_\psi) + k_{\text{FF}} \frac{\arctan(L\kappa_{\text{eff}})}{\delta_{\max}}, -1, 1\right), \tag{13}$$

$$\Delta\delta_{\max} = \frac{1.1\Delta t}{1 + 0.012 v_{\text{kmh}}}, \tag{14}$$

$$\delta_t = \delta_{t-1} + \text{clip}\big(\alpha(\delta_{\text{raw}} - \delta_{t-1}), -\Delta\delta_{\max}, \Delta\delta_{\max}\big). \tag{15}$$

True constant-time-headway (CTH) ACC ($d_{\text{lead}} < 38$ m, $v_{\text{lead}} \geq 8$ km/h, $d_0 = 8.0$ m, $\tau_h = 1.2$ s, $k_g = 0.85$ (km/h)/m $\equiv 0.236\,\text{s}^{-1}$) [17, 28, 30] follows:

$$d^*(v_{\text{ego,kmh}}) = d_0 + \tau_h \left(\frac{v_{\text{ego,kmh}}}{3.6}\right), \tag{16}$$

$$v_{\text{raw}} = v_{\text{lead}} + k_g\big(d_{\text{lead}} - d^*(v_{\text{ego,kmh}})\big), \tag{17}$$

$$v_{\text{ACC}} = \text{clip}\big(v_{\text{raw}},\, 0.0,\, \min(35.0,\, v_{\text{lead}} + 6.0)\big). \tag{18}$$

```
# src/control/shield_pid.py
d_star = 8.0 + 1.2 * (current_speed_kmh / 3.6)
acc_cap = min(35.0, lead_speed_kmh + 6.0)
raw_acc = lead_speed_kmh + 0.85 * (lead_distance_m - d_star)
acc_sync_speed = float(np.clip(raw_acc, 0.0, acc_cap))
traj_curv = (2.0 * ly) / (lx * lx + ly * ly)
eff_curv = 0.70 * (-road_curv * 0.045) + 0.30 * traj_curv
```

4 Experiments, Latency Breakdown, and Limitations

Closed-loop benchmarks were executed on an AMD Ryzen 5 7500F (6C/12T) and NVIDIA GeForce

Table 2: CARLA 0.9.16 (`Town10HD_Opt`) 10 paired episodes (680 m) and 10×5 cross-weather suite (3,400 m) under `vision_estimated` (Vis) vs. `gt` perception.

Scenario	Paired Weather	Event	10-Ep (Vis/GT)	50-Ep (Vis/GT)
FSD-1	ClearNoon	Red-light stop & go	1/1 / 1/1	5/5 / 5/5
FSD-2	CloudyNoon	Quintic left overtake	1/1 / 1/1	5/5 / 5/5
FSD-3	WetNoon	Crosswalk pedestrian AEB	1/1 / 1/1	5/5 / 5/5
FSD-4	SoftRainNoon	CTH-ACC (24 km/h, 16 m)	1/1 / 1/1	5/5 / 5/5
FSD-5	ClearSunset	Quintic right overtake	1/1 / 1/1	5/5 / 5/5
FSD-6	WetCloudyNoon	Lead stop-and-go recovery	1/1 / 1/1	5/5 / 5/5
FSD-7	MidRainyNoon	Front-lat latch (3/3) vs. rear approach (0/2 Vis)	1/1 / 1/1	3/5 / 4/5
FSD-8	WetSunset	Workzone right bypass	1/1 / 1/1	5/5 / 5/5
FSD-9	ClearNight	Night headlight ped AEB	1/1 / 1/1	4/5 / 5/5
FSD-10	HardRainNoon	Heavy-rain cornering ACC	1/1 / 1/1	5/5 / 5/5

RTX 4070 Ti (12 GB) workstation in MetaDrive 0.4.3 [21] ($\Delta t = 0.05$ s control step via `decision_repeat=5` over 0.01 s physics ticks; 415.3 steps / 20.77 s and 190.4 m per episode at $\bar{v} = 33.0$ km/h, totaling 4,153 steps / 1,904 m for 10 episodes and 20,765 steps / 9,520 m across $N = 50$ unseen maps) and live CARLA 0.9.16 [10] (`Town10HD_Opt` and unseen `Town03`, 20 Hz synchronous A* `GlobalRoutePlanner` [14]). Between MetaDrive and CARLA, the 196×392 crop/resize pipeline, frozen DINOv2+Laya weights, checkpoint `projector_best.pt`, temperatures $(0.05, 0.05, 0.10)$, CTH-ACC law, and 14-WP quintic spline are kept identical without retraining, whereas camera vertical FOV (90° CARLA vs. 65° MetaDrive) and map waypoint APIs (`GlobalRoutePlanner` vs. `navigation.checkpoints`) are adapted per simulator. Per-stage wall-clock latency measured on the same 1,981 frames with per-stage `torch.cuda.synchronize()` (`checkpoints/eval_metrics_carla.json`):


```
{"simulator": "carla", "ablation_mode": "full", "episodes": 10,
 "total_frames": 1981, "mean_latency_ms": 32.85,
 "mean_neural_latency_ms": 32.40, "base_stack_latency_ms": 31.67,
 "sim_tick_ipc_transport_ms": 6.40, "p95_latency_ms": 33.96,
 "stage_latency_breakdown_ms": {
   "preprocess_crop_resize": 0.42, "dinov2_vit_b14_frozen": 16.43,
   "spatial_projector_core_and_temporal": 0.82,
   "auxiliary_perception_head": 0.49,
   "laya_modernbert_and_discrete_heads": 13.55,
   "trajectory_head_6wp": 0.69,
   "safety_shield_bicycle_ff_pid": 0.45},
 "shield_interventions": {"rule1_emergency_brake": 46,
   "tp_hazard_aeb_events": 5, "fp_spurious_aeb_events": 1}}
```


All seven sub-stages sum exactly to 32.85 ms (32.40 ms neural forward pass + 0.45 ms shield/PID), and omitting `auxiliary_perception_head` (0.49 ms) and `trajectory_head_6wp` (0.69 ms) yields the exact 31.67 ms base latency (32.85 − 1.18 = 31.67 ms), while CARLA `world.tick()` sensor IPC adds 6.40 ms (39.25 ms total loop time < 50 ms budget). In Table 4 Part A, paired two-sided McNemar tests across the 50 matched episodes show statistically significant gains ($p < 0.01$) for DINOv2+Laya (47/50) over `mlp_head` (39/50, $p = 0.0078$), `expert_rule_policy` (38/50, $p = 0.0039$), `random_laya` (36/50, $p = 0.0010$), and `prompt_only` (33/50, $p = 0.0001$), whereas the 5-episode margin over the 2-layer trainable Transformer (42/50, discordant $b = 5, c = 0$, two-sided $p = 0.0625$, one-sided $p = 0.0313$) represents an observed advantage on rare night/rain hazards rather than a two-sided $\alpha = 0.05$ difference. Part B evaluates hybrid trajectory/controller ablations on the same 50-episode `vision_estimated` suite, showing that removing learned trajectory residuals ($\Delta\hat{\tau} = \mathbf{0}$, 45/50), removing the trajectory head ($\kappa_{\text{traj}} = 0$, 44/50), or removing map curvature ($\kappa_{\text{map}} = 0$, 41/50) each degrades completion, confirming that driving performance is achieved by the hybrid combination of learned visual inference, HD-map priors, and CTH-ACC/quintic control rather than the neural network alone.

References

[1] Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, Tianli Ding, Danny Driess, Avinava Dubey, Chelsea Finn, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. *arXiv preprint arXiv:2307.15818*, 2023.

[2] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9650–9660, 2021.

[3] Dian Chen, Brady Zhou, Vladlen Koltun, and Philipp Krähenbühl. Learning by cheating. In *Conference on Robot Learning (CoRL)*, pages 66–75, 2020.

[4] Kashyap Chitta, Aditya Prakash, Bernhard Jaeger, Zehao Yu, Katrin Renz, and Andreas Geiger. TransFuser: Imitation with transformer-based sensor fusion for autonomous driving. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 45(11):12878–12895, 2022.

[5] Felipe Codevilla, Matthias Müller, Antonio López, Vladlen Koltun, and Alexey Dosovitskiy. End-to-end driving via conditional imitation learning. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 4693–4700, 2018.

[6] ConvAI Innovations. Laya: Non-autoregressive low-latency decision models on bidirectional modernbert. https://huggingface.co/convaiinnovations/laya, 2024.

[7] R. Craig Coulter. Implementation of the pure pursuit path tracking algorithm. Technical Report CMU-RI-TR-92-01, Carnegie Mellon University, The Robotics Institute, 1992.

[8] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 4171–4186, 2019.

[9] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai,

Table 3: Closed-loop evaluation separating zero-actor-query `vision_estimated` perception from simulator `gt` state (`checkpoints/eval_metrics_carla.json`).

Simulator	Metric	Target KPI	`vision_estimated` Mode (No Sim Actor Query)	`gt` State Hybrid Mode	Status
CARLA 0.9.16	Single-Frame Latency (32.40 ms neural + 0.45 ms ctrl; +6.40 ms sim IPC)	≤ 40.0 ms	32.85 ms (30.4 FPS, P95: 33.96 ms)	32.85 ms (31.67 ms w/o aux+traj)	PASS
	`Town10HD_Opt`: 10 Paired Eps (680 m) / 50 Cross-Weather (3,400 m)	$\geq 85.0\%$	100.0% (10/10) / 94.0% (47/50, 3-seed 46.7 ± 0.6)	100.0% (10/10) / 98.0% (49/50)	PASS
	`Town10HD_Opt`: 10 Multi-Block Long Routes (200 m/ep, 2,000 m)	$\geq 85.0\%$	90.0% (9/10, $R_c = 95.4\%$)	100.0% (10/10, $R_c = 100.0\%$)	PASS
	`Town03` Unseen Map (10×200 m Roundabout/Grade, 2,000 m)	$\geq 80.0\%$	80.0% (8/10, $R_c = 91.2\%$, $\overline{\lvert e_y \rvert} = 0.112$ m)	90.0% (9/10, $R_c = 96.5\%$)	PASS
	Stress Test: 10% Frame Drop + 25% Patch Occlusion (50-Ep)	$\geq 80.0\%$	88.0% (44/50, ΔV_t + latch active)	94.0% (47/50)	PASS
	Teacher ECE_{mean} (ECE_b) / NLL / Brier / Physical Hazard FNR_{AEB}	≤ 0.0500	0.0142 (0.0195) / 0.089 / 0.028 / 1/48 evts (2.1%)	0.0000 (0.0000) / 0.048 / 0.014 / 0/48	PASS
	Aux Vision Lead Depth MAE / Cross-Track Error $\overline{\lvert e_y \rvert}$	≤ 5.0 m / ≤ 0.25 m	3.23 m / 0.094 m	3.23 m / 0.085 m	PASS
MetaDrive 0.4.3 ($\Delta t = 0.05$ s)	Single-Frame Latency ($\bar{v} = 33.0$ km/h, 190.4 m & 415.3 steps/ep)	≤ 40.0 ms	30.20 ms (33.1 FPS, P95: 31.67 ms)	30.20 ms (33.1 FPS)	PASS
	Arrival: 10-Ep (4,153 steps, 1,904 m) / 50-Ep (20,765 steps, 9,520 m)	$\geq 85.0\%$	100.0% (10/10) / 96.0% (48/50, CI [86.3, 99.5]%)	100.0% (10/10) / 100.0% (50/50; 98.7% arc)	PASS
	Unseen Closed-Loop Test ECE_{mean} / NLL / Brier ($M = 15$ Bins)	≤ 0.0500	0.0184 / 0.096 / 0.031 (Noisy Sensor)	0.0000 / 0.041 / 0.012 (GT Teacher)	PASS

Table 4: CARLA 0.9.16 architecture, baseline, and intra-policy latency ablations with 3-seed mean±std and paired two-sided McNemar p-values vs. `vision_estimated` DINOv2+Laya (Part A), plus hybrid trajectory/controller ablations evaluated on both 10-Ep `gt` and 50-Ep `vision_estimated` suites reporting event-level and tick-level TP/FP AEB (Part B; 10-Ep has 5 physical hazard events).

Part A: Decision & Backbone (Fixed Controller)	10-Ep	50-Ep Vis (3-Seed / McNemar p)	200 m	$\overline{\lvert e_y \rvert}$	$\lvert \Delta\delta \rvert_{\text{RMS}}$	TP/FP Evt (Ticks)
Keep-Lane Rule (`rule_only`, `gt`)	6/10	30/50 (60.0%, b=17, c=0, p=10^{-5})	5/10	0.142	0.0195	5/3 (44/14)
Telemetry-Prompt Only ($V_t = \mathbf{0}$, `vision_est`)	7/10	33/50 (32.7±0.6, b=14, c=0, p=0.0001)	6/10	0.154	0.0298	4/5 (38/26)
DINOv2 + Random Frozen Laya (395.21M rand, `vis`)	8/10	36/50 (35.7±1.2, b=11, c=0, p=0.0010)	6/10	0.136	0.0281	4/4 (41/21)
Direct `ExpertLabeler` Rules (`vision_estimated`)	8/10	38/50 (76.0%, b=9, c=0, p=0.0039)	7/10	0.131	0.0296	5/4 (43/18)
DINOv2 + 2-Layer MLP Head (No 395.21M Laya, `vis`)	8/10	39/50 (38.7±0.6, b=8, c=0, p=0.0078)	7/10	0.118	0.0265	5/3 (42/14)
Heuristic Overtake FSM (`heuristic_fsm`, `gt`)	8/10	39/50 (78.0%, b=8, c=0, p=0.0078)	7/10	0.126	0.0284	5/4 (44/20)
DINOv2 + 2-Layer Trainable Transformer (7.4M, `vis`)	9/10	42/50 (41.7±0.6, b=5, c=0, p=0.0625)	8/10	0.106	0.0249	5/2 (44/9)
DINOv2+Laya w/o Temporal ΔV_t (`no_temporal`, `vis`)	9/10	44/50 (43.7±0.6, b=3, c=0, p=0.2500)	8/10	0.102	0.0244	5/2 (44/8)
DINOv2+Laya + 100 ms Intra-Policy Lag (2-frame)	9/10	43/50 (86.0%, b=4, c=0, p=0.1250)	8/10	0.115	0.0258	5/2 (44/10)
DINOv2+Laya + 300 ms Intra-Policy Lag (6-frame)	7/10	34/50 (68.0%, b=13, c=0, p=0.0002)	6/10	0.168	0.0315	4/6 (39/33)
DINOv2+Laya (`vision_estimated`, Shield OFF)	9/10	43/50 (42.7±0.6, b=4, c=0, p=0.1250)	8/10	0.098	0.0241	4/1 (42/2)
DINOv2+Laya (`vision_estimated`, Shield ON)	10/10	47/50 (46.7±0.6, Reference)	9/10	0.094	0.0231	5/1 (45/4)
DINOv2+Laya (`gt` State, Shield OFF, `model_only`)	10/10	48/50 (47.7±0.6, b=0, c=1, p=1.0000)	9/10	0.086	0.0226	5/0 (44/1)
DINOv2+Laya (`gt` State, Shield ON, Full)	10/10	49/50 (48.7±0.6, b=0, c=2, p=0.5000)	10/10	0.085	0.0223	5/0 (44/2)
Part B: Hybrid Trajectory & Controller Ablation	**10-Ep GT**	**50-Ep Vis**	TTC_{min}	$J_{\dot{v},\text{RMS}}$	$\overline{\lvert e_y \rvert}$ (GT/Vis)	TP/FP Evt (Ticks)
Full Hybrid Stack ($0.70\kappa_{\text{map}} + 0.30\kappa_{\text{traj}}$)	10/10	47/50 (94.0%)	2.45 s	1.14	0.085/0.094	5/0 (44/2)
Kinematic Traj Prior Only ($\Delta\hat{\tau} = \mathbf{0}$, no $W_{\text{traj}}\hat{H}$)	10/10	45/50 (90.0%)	2.40 s	1.26	0.090/0.101	5/1 (44/4)
HD-Map Curvature Only ($\kappa_{\text{traj}} = 0$, no Traj Head)	10/10	44/50 (88.0%)	2.38 s	1.32	0.092/0.105	5/1 (44/4)
Learned Traj Curvature Only ($\kappa_{\text{map}} = 0, \kappa_{\text{eff}} = \kappa_{\text{traj}}$)	9/10	41/50 (82.0%)	2.15 s	1.74	0.118/0.124	5/2 (44/9)
− True CTH-ACC (Reverted to Fixed 16 m Gap)	10/10	42/50 (84.0%)	1.42 s	1.48	0.095/0.108	5/14 (46/66)
− 28 m Spatial Quintic Spline (3.5 m Step WPs)	10/10	41/50 (82.0%)	1.95 s	3.65	0.148/0.162	5/10 (45/49)
− Speed-Damped Linear Steer Ramp (Unramped PID)	10/10	39/50 (78.0%)	1.82 s	4.41	0.184/0.198	5/15 (45/73)
All Removed (Discrete 2 m Step WPs + Fixed-Gap ACC)	10/10	35/50 (70.0%)	1.18 s	4.82	0.218/0.236	5/19 (46/92)

Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.

[10] Alexey Dosovitskiy, German Ros, Felipe Codevilla, Antonio Lopez, and Vladlen Koltun. CARLA: An open urban driving simulator. In *Proceedings of the 1st Annual Conference on Robot Learning (CoRL)*, pages 1–16, 2017.

[11] Marius Dupuis, Martin Strobl, and Hans Grezlikowski. OpenDRIVE format specification, rev. 1.4. *VIRES Simulationstechnologie GmbH*, 2010.

[12] Tamar Flash and Neville Hogan. The coordination of arm movements: An experimentally confirmed mathematical model. *Journal of Neuroscience*, 5(7):1688–1703, 1985.

[13] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning (ICML)*, pages 1321–1330, 2017.

[14] Peter E. Hart, Nils J. Nilsson, and Bertram Raphael. A formal basis for the heuristic determination of minimum cost paths. *IEEE Transactions on Systems Science and Cybernetics*, 4(2):100–107, 1968.

[15] Gabriel M. Hoffmann, Claire J. Tomlin, Michael Montemerlo, and Sebastian Thrun. Autonomous automobile trajectory tracking for off-road driving: Controller design, experimental validation and racing. *Proceedings of the American Control Conference (ACC)*, pages 2296–2301, 2007.

[16] Yihan Hu, Jiazhi Yang, Li Chen, Keyu Li, Chonghao Sima, Xizhou Zhu, Siqi Chai, Senyao Du, Tianwei Lin, Wenhai Wang, et al. Planning-oriented autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 17853–17862, 2023.

[17] Petros A. Ioannou and C. C. Chien. Autonomous intelligent cruise control. *IEEE Transactions on Vehicular Technology*, 42(4):657–672, 1993.

[18] Bernhard Jaeger, Kashyap Chitta, and Andreas Geiger. Hidden biases of end-to-end driving models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 8240–8249, 2023.

[19] Bo Jiang, Shaoyu Chen, Qing Xu, Bencheng Liao, Jiajie Chen, Helong Zhou, Qian Zhang, Wenyu Liu, Chang Huang, and Xinggang Wang. VAD: Vectorized scene representation for efficient autonomous driving. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 8340–8350, 2023.

[20] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, et al. OpenVLA: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024.

[21] Quanyi Li, Zhenghao Peng, Lan Feng, Qihang Zhang, Zhenghai Xue, and Bolei Zhou. MetaDrive: Composing diverse driving scenarios for generalizable reinforcement learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 45(3):3461–3475, 2022.

[22] Zhiqi Li, Wenhai Wang, Hongyang Li, Enze Xie, Chonghao Sima, Tong Lu, Yu Qiao, and Jifeng Dai. BEVFormer: Learning bird's-eye-view representation

from multi-camera images via spatiotemporal transformers. In *European Conference on Computer Vision (ECCV)*, pages 1–18, 2022.

[23] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

[24] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

[25] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research (TMLR)*, 2024.

[26] Brian Paden, Michal Čáp, Sze Zheng Yong, Dmitry Yershov, and Emilio Frazzoli. A survey of motion planning and control techniques for self-driving urban vehicles. *IEEE Transactions on Intelligent Vehicles*, 1(1):33–55, 2016.

[27] Aditya Prakash, Kashyap Chitta, and Andreas Geiger. Multi-modal fusion transformer for end-to-end autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7077–7087, 2021.

[28] Rajesh Rajamani and Chuan Zhu. An experimental comparative study of autonomousogh and cooperative vehicle-following control laws. *IEEE Transactions on Control Systems Technology*, 10(3):371–383, 2002.

[29] Sebastian Thrun, Michael Montemerlo, Hendrik Dahlkamp, David Stavens, Andrei Aron, James Diebel, Philip Fong, John Gale, Morgan Halpenny, Gabriel Hoffmann, et al. Stanley: The robot that won the DARPA grand challenge. *Journal of Field Robotics*, 23(9):661–692, 2006.

[30] Martin Treiber, Ansgar Hennecke, and Dirk Helbing. Congested traffic states in empirical observations and microscopic simulations. *Physical Review E*, 62(2):1805–1824, 2000.

[31] Benjamin Warner, Antoine Chaffin, Benjamin Clavié, Orion Weller, Oskar Hallström, Said Taghadouini, Alexis Gallagher, Raja Biswas, Faisal Ladhak, Tom Aarsen, et al. Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference. *arXiv preprint arXiv:2412.13663*, 2024.

[32] Moritz Werling, Julius Ziegler, Sören Kammel, and Sebastian Thrun. Optimal trajectory generation for dynamic street scenarios in a Frenet frame. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 987–993, 2010.